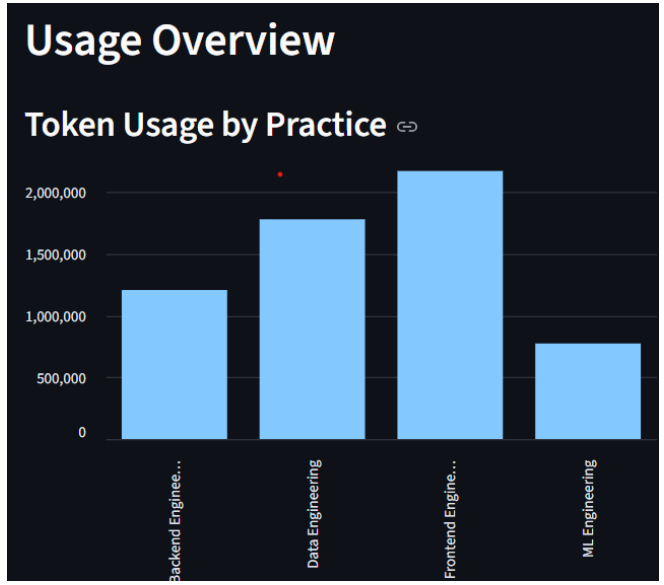
Claude Code Analytics Platform

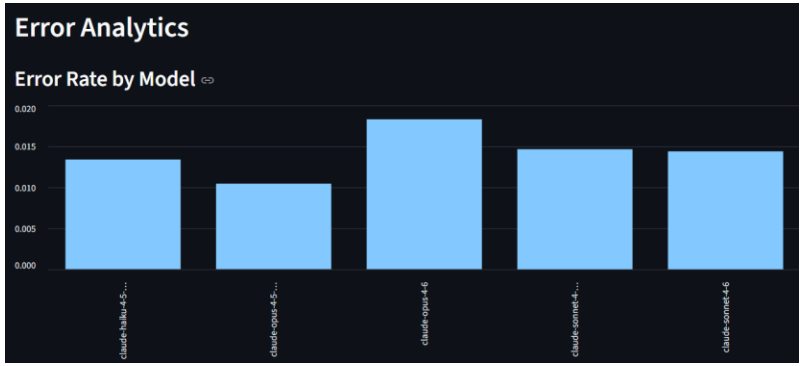
- **Goal:** Analyze Claude Code telemetry to find developer patterns
- **Dataset:** 100 engineers, 5000 sessions, 2 months
- **Tech:** Python, Pandas, Streamlit, JSON/CSV, SQL
- **LLM usage:** ChatGPT & Claude for script generation & validation

Streamlit

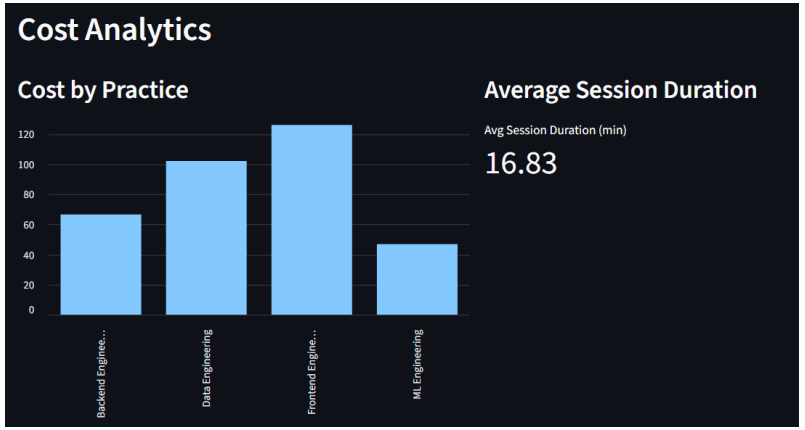
Usage Metrics & Patterns

- Token usage per practice
- Token Consumption by practise
- Avg session duration
- API success/error rate



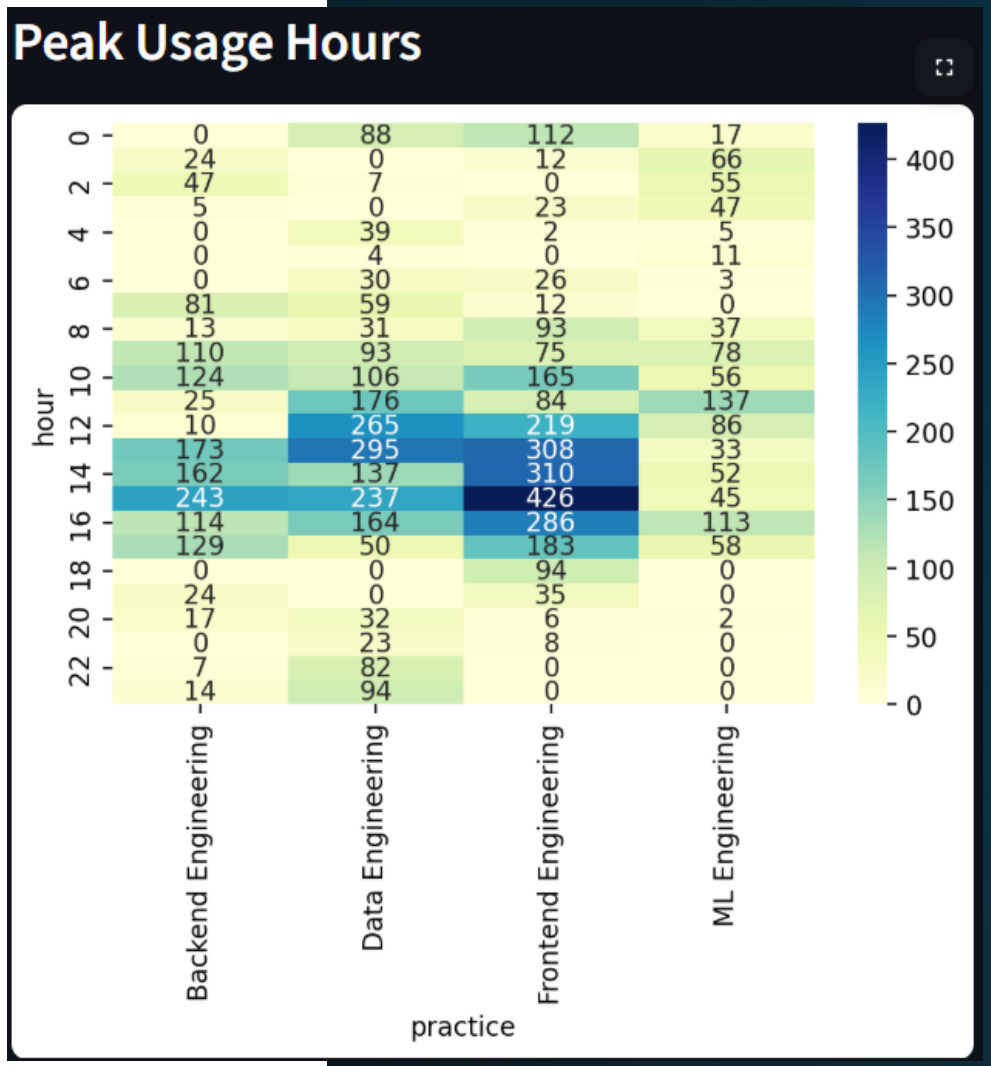


	tool_name	success_rate
0	AskUserQuestion	1
3	ExitPlanMode	1
8	Task	1
4	Glob	0.995
10	TaskUpdate	0.9938
15	Write	0.9919
9	TaskCreate	0.9885
2	Edit	0.9853
11	TodoWrite	0.9852
5	Grep	0.9849



Peak Usage Hours & Trends

- Most active hours: 12:00–16:00, with a clear peak around 15:00
- Workday activity pattern: Usage is strongly aligned with standard working hours.
- Insights: Frontend Engineering shows the highest single-hour peak (15:00).
  Data Engineering demonstrates consistently high activity during core working hours (12:00–16:00).
  Usage significantly decreases after 18:00 across all practices.

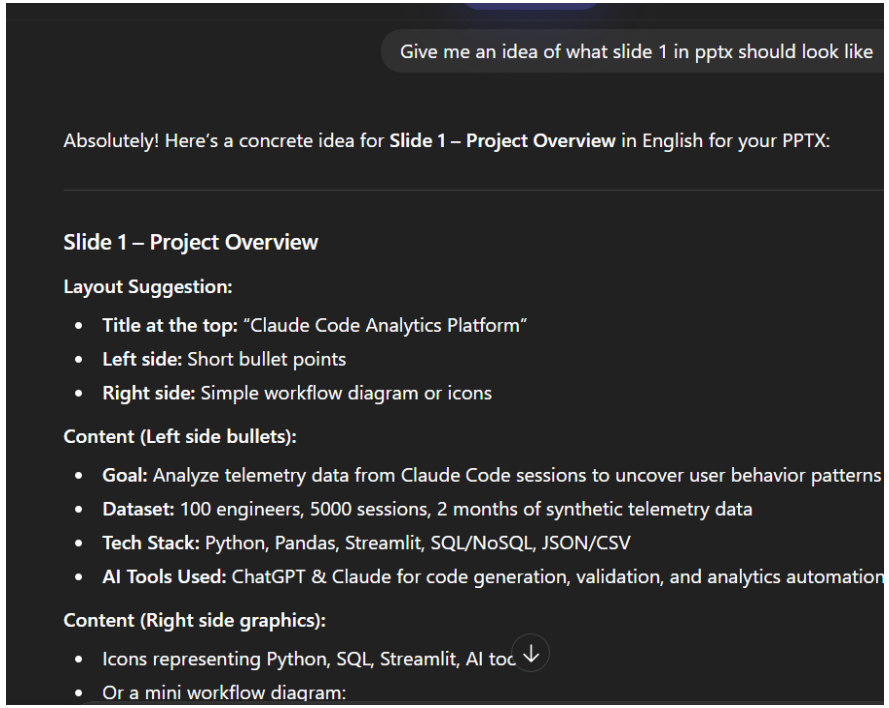


Key Insights & Recommendations

- Insights
- Backend engineers use Claude Code most often in the middle of the working day
- Tool decision accept rate is >95%, but WebFetch sometimes fails
- Sessions with more than 3 API requests are rare, but expensive
- Recommendations:
- Optimize WebFetch requests
- Focus user education on peak usage hours
- Implement caching for ML engineers

LLM Tools & Workflow

- Tools used:ChatGPT,Github Copilot Chat
- Example prompt: Idea to generate slide content for a PowerPoint presentation
- Result: Generated text content ready to be placed on slides



Give me an idea of what slide 1 in pptx should look like

Absolutely! Here's a concrete idea for **Slide 1 – Project Overview** in English for your PPTX:

Slide 1 – Project Overview

Layout Suggestion:

- **Title at the top:** "Claude Code Analytics Platform"
- **Left side:** Short bullet points
- **Right side:** Simple workflow diagram or icons

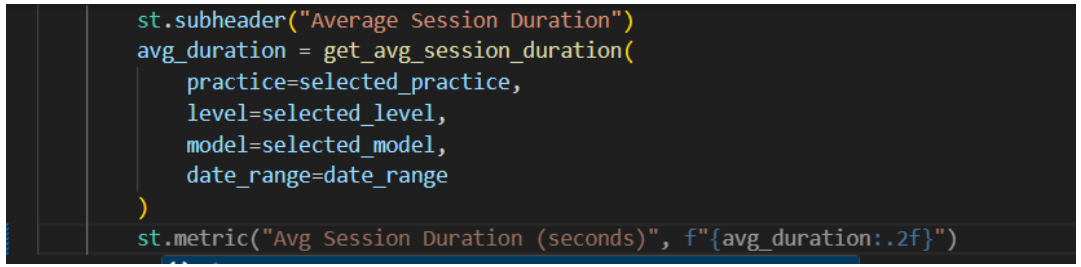
Content (Left side bullets):

- **Goal:** Analyze telemetry data from Claude Code sessions to uncover user behavior patterns
- **Dataset:** 100 engineers, 5000 sessions, 2 months of synthetic telemetry data
- **Tech Stack:** Python, Pandas, Streamlit, SQL/NoSQL, JSON/CSV
- **AI Tools Used:** ChatGPT & Claude for code generation, validation, and analytics automation

Content (Right side graphics):

- Icons representing Python, SQL, Streamlit, AI too
- Or a mini workflow diagram:

- Example copilot: displaying average session duration result
- Result: Generated plot and formatted table ready for dashboard integration



```
st.subheader("Average Session Duration")
avg_duration = get_avg_session_duration(
    practice=selected_practice,
    level=selected_level,
    model=selected_model,
    date_range=date_range
)
st.metric("Avg Session Duration (seconds)", f"{avg_duration:.2f}")
```